TABLE I
CLASSIFICATION ACCURACY AND NETWORK SIZE ON FIVE TEMPORAL BENCHMARKS.

Dataset	Method	Network	Parameters (K)	Accuracy (%)
S-MNIST	GLIF (Yao et al. [31])	Rec	150	96.64
	ASRNN (Yin et al. [1])	Rec	150	98.70
	DH-SNN (Zheng et al. [9])	Rec	80	98.90
	BRF (Higuchi et al. [6])	Rec	69	99.10
	TC-LIF (Zhang et al. [8])	Rec	150	99.20
	DMP-SNN (II) (Sun et al. [11])	DMP	73	99.20
	SPRiF (ours)	**Rec**	**67**	$\mathbf{99.28 \pm 0.06}$
PS-MNIST	GLIF (Yao et al. [31])	Rec	150	90.47
	ASRNN (Yin et al. [1])	Rec	150	94.30
	DH-SNN (Zheng et al. [9])	Rec	80	94.52
	BRF (Higuchi et al. [6])	Rec	69	95.20
	TC-LIF (Zhang et al. [8])	Rec	150	95.36
	DMP-SNN (I) (Sun et al. [11])	DMP	61	95.50
	TC-LIF^{R} (Zhang et al. [8])	Rec	150	95.10 $\pm$0.33
	BRF^{R} (Higuchi et al. [6])	Rec	69	95.02 $\pm$0.41
	DMP-SNN^{N} (Sun et al. [11])	DMP	67	95.41 $\pm$0.52
	SPRiF (ours)	**Rec**	**67**	$\mathbf{95.86 \pm 0.14}$
QTDB	ASRNN (Yin et al. [1])	Rec	1.8	85.90
	DH-SNN (Zheng et al. [9])	Rec	1.78	86.35
	BRF (Higuchi et al. [6])	Rec	1.73	87.00
	SE-adLIF (Baronig et al. [2])	Rec	1.8	86.88
	SPRiF (ours)	**Rec**	**1.7**	$\mathbf{88.43 \pm 0.36}$
GSC	SNN-SFA (Salaj et al. [32])	Rec	4190	91.21
	ASRNN (Yin et al. [1])	Rec	310	92.10
	RadLIF (Bittar and Garner [33])	Rec	1200	94.51
	DH-SNN (Zheng et al. [9])	Rec	130	93.80
	TC-LIF (Zhang et al. [8])	Rec	200	94.84
	SPRiF (ours)	**Rec**	**130**	$\mathbf{94.55 \pm 0.31}$
SHD	Heterogeneous SNN (Perez-Nieves et al. [3])	Rec	110	82.50
	ASRNN (Yin et al. [1])	Rec	140	90.40
	DH-SNN (Zheng et al. [9])	Rec	50	91.34
	BRF (Higuchi et al. [6])	Rec	110	91.70
	TC-LIF (Zhang et al. [8])	Rec	140	88.91
	MPS-SNN (Ding et al. [34])	Rec	390	91.19
	DGN (Bai et al. [35])	Rec	220	87.78
	TC-LIF^{R} (Zhang et al. [8])	Rec	140	88.73 $\pm$0.44
	BRF^{R} (Higuchi et al. [6])	Rec	110	91.28 $\pm$0.53
	SPRiF (ours)	**Rec**	**50**	$\mathbf{91.66 \pm 0.42}$

Notes: K denotes 10^3 network parameters. Rec: recurrent network; DMP: dual-memory-pathway architecture. Unmarked baseline rows are published results under their original protocols; ^{R}our reruns; ^{N}our run of the full DMP-SNN architecture. SPRiF rows are bold. Our results report mean $\pm$ sample SD over five seeds. GSC reports best validation accuracy; the other four tasks report best test accuracy across epochs.